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Estimating growth within size-structured fishery stock assessments: What is the state of the art and what does the future look like?

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ABSTRACT

Most growth studies have considered the relationship between age and growth. Such relationships are essential to age-structured fishery stock assessments. In contrast, assessments based on stage-structured population dynamics models require information on the probability of animals moving from one stage to each of the other stages at each time-step. Size-structured population dynamics models are a special case of stage-structured population dynamics models in which each class represents a unique set of sizes. These models form the basis of assessments for many valuable, hard to age, marine species, including crabs, abalone, lobsters and prawns. Growth within these models is governed by a size-transition matrix. The values for the parameters of a size-transition matrix can either be estimated externally to the assessment, generally utilizing data from tag-recapture experiments, or the estimation of growth can be integrated within the assessment model. This paper reviews the approaches used to construct size-transition matrices, including the underlying structural formulation, statistical estimation framework, and the consequences of error when specifying these matrices on the ability to estimate population size and manage populations sustainably.

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1. Introduction

Management advice for many of the world's fish and invertebrate stocks is based on quantitative stock assessments that provide estimates of productivity, and time-series of biomass and recruitment. These stock assessments often form the basis for forecasts and can serve as the basis for the application of harvest control rules. Single-species stock assessments can be classified into three broad classes: (a) surplus production (or biomass dynamic) models, (b) age-structured models, and (c) size- (or length-/stage-) structured models (hereafter referred to as size-structured models). The majority of contemporary stock assessments are based on age-structured models. However, such methods cannot easily be applied to species which are hard to age such as crustaceans and molluscs (in particular crabs, rock lobsters, prawns, abalone, and oysters), but also some teleosts such as tunas and billfish, all of which form the basis for very valuable fisheries.

Stock assessments may be based on stage-structured models when (a) there are no data on the age-structure of the population or the removals from the population, (b) many of the key biological processes (such as growth, selectivity, natural mortality, and fecundity) are size- rather age-based, or (c) there is concern regarding the impact size-selective mortality on the distribution of length-at-age. Size-structured population dynamics models are a special case of stage-structured population dynamics models in which each stage represents a unique set of sizes. Assessments based on size-structured population dynamics models require information on the probability of animals growing from one size-class to each of the other size-classes at each time-step and staying in the same size class.

Punt et al. (2013a) provide a review of methods based on size-structured population dynamics models which can utilize size-composition data to estimate biomass, recruitment and fishing mortality. Most of the recent methods are based on 'integrated' (or 'statistical') approaches. These methods involve modelling the population dynamics and how the observations pertain to the model separately. The earliest such methods were developed by Fournier

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and Archibald (1982) in the context of age-structured dynamics, while the first integrated size-structured assessment method was developed by Sullivan et al. (1990). There are now many variants on the basic approach of Sullivan et al. (1990).

The basic dynamics of most integrated size-structured stock assessment methods are governed by the equation:

$$N_{t+1} = \mathbf{X}_t \mathbf{S}_t N_t + \phi R_{t+1} \quad (1)$$

where N_t is a column vector of length H (number of size-classes, which need not all be the same size) containing the numbers-at-size at the start of time-step t, $\mathbf{X}_t$ is the transition matrix (dimension $H \times H$) for time-step t (where $X_{i,j,t}$ is the probability of growing from size-class j to size-class i during time-step t), $\mathbf{S}_t$ is a $H \times H$ diagonal matrix, with the diagonal terms equal to the proportion surviving (the complement of fishing and natural mortality) at size during time-step t, R_t is the recruitment in numbers during time-step t (assumed to occur at the end of the time-step after mortality and growth) and ϕ is a column vector of length H containing the proportions of recruitment to each size-class.

Growth appears in Eq. (1) through the matrix $\boldsymbol{X}$ and the vector ϕ. The focus for this paper is on the matrix $\boldsymbol{X}$ because there are seldom data to allow ϕ to be estimated (some assessments, e.g. Zheng and Siddeek (2013) do, however, estimate it). Consequently, the values for ϕ are usually pre-specified, and the smallest population size-class selected so that ϕ covers the full size distribution of cohorts entering the fished population in what are assumed to be the correct proportions by size-class. An alternative is to start the model at very small sizes in which case the values of ϕ will be inconsequential for the estimates of quantities of management interest. However, there is likely little benefit to starting a model at very small size unless there are data on growth of such animals, which is often not the case.

Size-transition matrices are used to model both crustacean growth and the continuous growth of molluscs and, less often, fish. The moulting growth of crustaceans (reviewed by Caddy, 1987) has been modelled by separately considering the discrete increase in body size, moult increment (Hiatt, 1948; McGarvey and Feenstra, 2001), and time period between moulting events, the intermoult period (Hoenig and Restrepo, 1989; Zheng et al., 1995; Millar and Hoenig, 1997). A focus of crustacean growth studies had been to quantify the slowing of growth following sexual maturity, particularly for females (Hiatt, 1948; Kurata, 1962; Mauchline, 1977; Somerton, 1980; Easton and Misra, 1988). The size-transition growth model formulation can describe both continuous and moulting growth because it applies to populations, not to individuals. The probabilities of transition are understood to predict the proportions of animals growing to larger size classes (or remaining in their size class) over time spanning one or several model time steps.

While it is theoretically possible to estimate the individual elements of size-transition matrices as independent parameters, in practice these are too numerous. Instead, size-transition probabilities are parameterized as functions of (pre-growth) body length, often with different growth matrices by season and sex. The growth increment is usually modelled using the von Bertalanffy equation, although dependence on other factors affecting growth, such as population density and state of sexual maturity are also possible, as described below.

For the purposes of this paper, the matrix $\boldsymbol{X}$ is defined as the probability of, or proportion, of animals growing from one size-class to each of the other size-classes (usually constrained so that shrinkage is prohibited). Some assessments generalize the N vector to include maturity state and shell condition (e.g. Stockhausen et al., 2013; Turnock and Rugolo, 2013; Zheng and Siddeek, 2013) in which case $\boldsymbol{X}$, now a multi-dimensional array, represents movement among states as well as growth. In its simplest form, the matrix $\boldsymbol{X}$ assumes that all animals grow each time-step, i.e. there is no terminal moult[i] for crabs and no intermoult intervals longer or shorter than the modelled time-step. Allowing for different states permits, for example, terminal moults to be represented in the model. Eq. (1) is written in terms of time-steps, which may be a year or a shorter time-period, and each time-step need not be of the same duration. For example, the assessment for rock lobsters, *Jasus edwardsii*, conducted by Punt and Kennedy (1997) was based on a monthly time-step and growth occurred at the end of March, June, September and December rather that during each month, primarily due to lack of tag-recovery data to estimate monthly size-transition matrices.

The parameter values for the size-transition matrix can either be estimated independently of the assessment or integrated within the assessment. The latter allows data other than the tag-recapture information to impact estimates of growth, including size frequencies sampled from the catch, and allows uncertainty in the estimation of the size-transition matrix to be accounted for more rigorously in final stock assessment outputs. However, integrating growth estimation into the assessment is more computationally intensive, and estimation outside of the assessments has been preferred by some authors (e.g. Chen et al., 2000). The main focus for this paper is on assessments for crustaceans and molluscs because these are the species groups to which size-structured assessment methods have been applied most frequently. The example applications are taken from Australia, New Zealand, South Africa and Alaska as these are the regions where size-structured 'integrated' methods of assessment have been applied most often. The methods of this paper are framed in the context of size-structured models. However, they can also be used in models in which the population is represented using an age- and size-structured model where each cohort is subdivided into pre-specified size classes (e.g. Frøysa et al., 2002; Gilbert et al., 2006; Gardner et al., 2007) or where length bins grow with the cohort length-at-age distribution (McGarvey et al., 2007). Section 2 of this paper identifies a set of models on which size-transition matrices can be based and how the values for their parameters can be estimated using tag-recapture data. Section 3 outlines how the values for size-transition matrices can be estimated when that estimation is fully integrated into the assessment. Section 4 describes how errors in the size-transition matrix can impact the quantities on which management advice are based. Section 5 summarises the current state of progress with the use of size-transition matrices. Finally, Section 6 outlines some general issues and provides recommendations and suggestions for future work.

2. Methods to estimate size-transition matrices

The growth process for crustaceans can be characterized by a combination of the distribution for the time between moults and the distribution for the growth increment given a moult (Caddy, 1987; Chang et al., 2012). For a discrete-time discrete-size-class model such as Eq. (1), this can be approximated by a combination of the probability of a growth transition (by size) multiplied by the distribution for the growth increment (by size) given a moult occurs. Many analyses have been based on explicitly modelling growth increment and the probability of moulting or equivalently the intermoult period (e.g. Chen and Kennedy, 1999; Ehrhardt, 2008; Zheng and Siddeek, 2013; Turnock and Rugolo, 2013).

[i] A terminal moult implies that at some stage (usually after reaching sexual maturity) the animals stops growing.

Denoting the probability of moulting as a function of pre-moult size class i as P_i and the matrix which defines the growth increment as Y, the matrix X is defined as:

$$X = \begin{pmatrix} (1-P_1)+P_1Y_{1,1} & 0 & 0 & 0 \\ P_1Y_{1,2} & (1-P_2)+P_2Y_{2,2} & . & 0 \\ . & . & . & . \\ P_1Y_{1,H} & P_2Y_{2,H} & . & 1 \end{pmatrix} \quad (2)$$

for the case in which no shrinkage is possible (Zheng et al., 1995; Moser et al., 2002). Note that in the absence of data on whether animals moulted, the probability of not moulting and of moults that do not lead to animals growing out of their current size-class are confounded. Consequently, most authors do not distinguish between not moulting and moulting but staying in the original size-class, i.e.:

$$X = \begin{pmatrix} Y_{1,1} & 0 & 0 & 0 \\ Y_{1,2} & Y_{2,2} & . & 0 \\ . & . & . & . \\ Y_{1,H} & Y_{2,H} & . & 1 \end{pmatrix} \quad (3)$$

The values for the $Y_{i,j}$ are generally based on model output, often via continuous probability density functions whose parameters are expressed as functions of pre-growth body size. Consequently, upper-tail entries such as $Y_{1,H}$ may be small but will not be zero. It is therefore not uncommon (e.g. Turnock and Rugolo, 2013) to restrict growth to a limited number of size-classes and to assume that growth above this size class will be zero. The number of size classes to use could be derived based on physiological considerations.

Tag-recaptures are the key data source used to estimate size-transition matrices. In principle, direct measurements of moult duration or moult probability and growth increment given size can be obtained from captive animals. However, unlike in-situ experiments, it is not clear whether the data on growth from captive animals are representative of animals in the wild. Although in-situ tag-recapture experiments should provide data pertinent for estimating size-transition matrices, these experiments may be inappropriate if (a) tagging impacts growth rates (e.g. Lloyd-Jones et al., 2014a), or (b) the tagged animals are not representative of the entire population of interest. The data from a tagging experiment are triplets $\{l_{1,i}, l_{2,i}, \Delta t_i\}$ where $l_{1,i}$ and $l_{2,i}$ are lengths at tagging and recapture and Δt_i is the time-at-liberty.

The ideal (non-parametric) basis to estimate the matrix $\boldsymbol{X}$ would be to tag a very large number of animals over a wide range of sizes, and recapture them exactly a year after release (the 'anniversary method': Hancock and Edwards, 1967; Bennett, 1974). The size-transition matrix could then be defined empirically. However, it is seldom possible to tag sufficient numbers of animals to apply this method, and recaptures in reality will occur at random times over several years. The methods used in practice for estimating size-transition matrices can generally be divided into (a) those which estimate the mean growth curve and variance about that curve and assume that differences between observed and model-predicted recapture sizes are due to measurement or process error, and (b) those which allow for individual variation in growth parameters. An exception to these two classes of method is the approach used to develop a size-transition matrix for American lobster (*Homarus americanus*) which is based on an individual-based simulation model (ASMFC, 2009). Most methods estimate growth curves, while a few methods are specifically designed to estimate the size-transition matrix directly. Given a model-estimated or growth-curve mean growth increment and the variance of growth increment as functions of premoult size, the elements of the growth matrix are computed by integrating over a probability density function (pdf) or otherwise assigning probabilities of transition based on the predicted growth.

A key structural assumption of both types of methods is the growth curve used. When it is integrated into an assessment, the probability of moulting has usually been assumed to be a declining logistic function, i.e.:

$$P_i = \left(1+\exp[-A+B\bar{\ell}_i]\right)^{-1} \quad (4)$$

where A, and B are the parameters of the relationship, and $\bar{\ell}_i$ is the mid-point of size-class i (the bounds of size-class i are denoted (ℓ_i^-, ℓ_i^+)). In contrast, various functional forms have been assumed when modelling the growth increment given premoult size and time, $I(\ell, t, \delta)$ where ℓ is premoult size, t is time and δ is a vector of parameters. Chang et al. (2012) provide a comprehensive overview of the models used to define the expected growth increment as a function of premoult size. Most studies on which assessments are based have taken the function $I()$ to be the von Bertalanffy curve, where the growth increment is a linear function of premoult size. However, many assessments in Australia and New Zealand have used the more general Schnute growth model (Schnute, 1981; Francis, 1995; Punt et al., 1997; Haist et al., 2009):

$$I\left(\ell, \Delta t, \gamma, \kappa, \xi\right) = \left\{\ell^\gamma \exp\left(-\kappa\Delta t\right) + \xi^\gamma\left(1-\exp\left(-\kappa\Delta t\right)\right)\right\}^{1/\gamma} \quad (5)$$

Several common growth curves, including the logistic, von Bertalanffy, and Gompertz are special cases of Eq. (5). The parameters of Eq. (5) can be transformed into parameters which are biologically meaningful using the transformation identified by Haist et al. (2009), i.e.:

$$\kappa = -\ln\left(1-\frac{a-b}{\ell_1^\gamma - \ell_2^\gamma}\right); \quad \xi = \frac{\ell_1^\gamma - \ell_1^\gamma\left(b/a\right)}{1-\left(b/a\right)} \quad (6)$$

where $a = \left(\ell_1+\lambda_1\right)^\gamma - \ell_1^\gamma$ and $b = \left(\ell_2+\lambda_2\right)^\gamma - \ell_2^\gamma$, ℓ_1, ℓ_2 are two pre-specified sizes, and λ_1, λ_2 are the expected growth increments of animals of sizes ℓ_1 and ℓ_2. This choice of parameterization should reduce the correlation among the parameters (Francis, 1988).

Other growth curves which have been considered in assessments include generalizations of the von Bertalanffy growth curve which allow for seasonal variation in the growth rate (Pauly and Gaschüz, 1979; Francis, 1988; Pauly et al., 1992), and the polynomial formulation of McGarvey and Feenstra (2001). Surette and Allard (2008) and Turnock and Rugolo (2013) addressed slowing of growth at the onset of maturity by assuming the growth increment to be a piecewise linear model of premoult size, extending broken stick models (Hiatt, 1948; Kurata, 1962) to a size-transition formulation.

2.1. General structure of size-transition matrices

The general form for the entries in the size-transition matrix is:

$$X_{i,j} = \int_{\ell_j^-}^{\ell_j^+}\int_{\ell_i^-}^{\ell_i^+} F\left(\delta, k, l\right) \mathrm{d}l\mathrm{d}k \quad (7)$$

where l and k are sizes, F is some pre-specified function, and δ is a vector of parameters. Few authors (e.g. Bardos, 2005) have implemented Eq. (7) exactly owing to its computational demands (and the need to know the pdf for lengths within size-class j). Consequently, most authors have followed Punt et al. (1997) and evaluated Eq. (7) under the assumption that all animals in the

initial size-class j have size equal to the mid-point of size-class j, $\bar{\ell}_j$, i.e.:

$$X_{i,j} = \int_{\ell_i^-}^{\ell_i^+} F\left(\delta, \bar{\ell}_j, l\right) \mathrm{d}l. \tag{8}$$

The function F defines the probability distribution for the growth increment given premoult length. This pdf is usually normal, lognormal or gamma, but Bergh and Johnston (1992) used a beta distribution, which can be skewed to the left or right. The integrals in Eqs. (7) and (8) have generally been evaluated numerically (e.g. Bardos, 2005). However, it may be more computationally convenient to use Monte Carlo methods in complex cases, for example, when both the asymptotic size and growth rate vary among individuals (e.g. Chen et al., 2003).

2.2. Modelling growth increments directly

The aim of this approach (and that of the next approach) is to estimate the values for the parameters of the function F in Eqs. (7) and (8). It does so by postulating that each animal follows the same underlying growth curve and that deviations about that growth curve are due to observation or process error. Modelling the relationship between pre- and post-growth size involves either regressing the observed growth increments $\left(\Delta l_i = l_{2,i} - l_{1,i}\right)$ on initial size and time-at-liberty (Eq. (9a)) or discretising the release and recapture sizes and the time-at-liberty and computing the individual elements of the size-transition matrix, that is the probability of observing an animal which grows from the release size-class to the recapture size-class given it was at liberty for a certain number of time-steps (Eq. (9b)):

$$\ell nL = \sum_i f\left(\Delta l_i | l_{1,i}, \Delta t_i, \delta\right) \tag{9a}$$

$$\ell nL = \sum_i \ell n[X^{\Delta t_i}]_{\tilde{l}_{1,i}, \tilde{l}_{2,i}} \tag{9b}$$

where $\tilde{l}_{1,i}$ is the size-class within which $l_{1,i}$ falls, $\tilde{l}_{2,i}$ is the size-class within which $l_{2,i}$ falls, and Δt_i is the number of time-steps over which animal i was at liberty (Punt et al., 1997; Moser et al., 2002). The function f defines the pdf for the growth increment observations. Typical choices for this function are (a) the normal, (b) the gamma, and (c) the lognormal:

$$f\left(\Delta l_i | l_{1,i}, \Delta t_i, \delta\right) = \frac{1}{\sqrt{2\pi}\,\sigma_i^2} \exp\left(-\frac{\left(\Delta l_i - I\left(l_{1,i}, \Delta t_i, \delta\right)\right)^2}{2\,\sigma_i^2}\right) \tag{10a}$$

$$f\left(\Delta l_i | l_{1,i}, \Delta t_i, \delta\right) = \frac{\exp\left(-\gamma_i \Delta l_i\right)\left(l - \Delta l_i\right)^{r_i - 1}}{\Gamma\left(r_i\right)\left(\gamma_i\right)^{-r_i}} \tag{10b}$$

$$f\left(\Delta l_i | l_{1,i}, \Delta t_i, \delta\right) = \frac{1}{\sqrt{2\pi}\,\sigma_i^2\,\Delta l_i} \exp\left(-\frac{\left(\ell n \Delta l_i - \ell n\left(I\left(l_{1,i}, \Delta t_i, \delta\right)\right)\right)^2}{2\,\sigma_i^2}\right) \tag{10c}$$

where $r_i = 1/CV_i^2$ and $\gamma_i = 1/\left(I\left(l_{1,i}, \Delta t_i, \delta\right)\ CV_i^2\right)$. σ_i and CV_i in Eq. (10) quantify the extent of variation in growth increment for animal i. Several options exist for defining the variance function, including that it is constant, proportional to the premoult size, proportional to the time-at-liberty (Francis, 1988), a polynomial function of premoult size (McGarvey and Feenstra, 2001), or proportional to the growth increment with a minimum standard error (Haist et al., 2009).

Surette and Allard (2008) developed an estimation method which allowed for maturity to occur over a range of sizes and hence the distribution of recapture lengths for a tag-recapture experiment to be a mixture distribution given distributions of growth increment by maturity state and the logistic probability of being mature as a function of length.

Francis (1988) extended the basic approach for estimating growth increment as a function of pre-growth size, time-at-liberty, and time of year to allow for measurement error and the probability of outlying estimates, i.e.:

$$\ell nL = \sum_i \ell n\left[(1-p)\,\ell n\ \ f\left(\Delta l_i | l_{1,i}, \Delta l_i, \delta\right) + \frac{p}{R}\right] \tag{11}$$

where the variance of the difference between the observed and model-predicted growth increments consists of the sum of the process error in growth increment, σ_i^2, and the variance of the measurement error, τ^2. p in Eq. (11) is the probability of an outlier and R is the range (largest minus smallest) of observed growth increments. The final term of Eq. (11) attempts to reduce the impact of outlying observations in tag-recapture data. Maunder (2002) further extended the basic approach to allow the variability in recapture size to depend on length-at-release and time-at-liberty, and used this formulation in a size-structured assessment of skipjack tuna (Maunder, 2012). Maunder (2002) also provided a model structure in which there is error measuring animals both at release and at recapture.

Use of Eq. (9b) involves defining a mean growth function and a model for the variation about mean growth and then applying Eq. (7) or (8) (Punt et al., 1997, 2009).

2.3. Modelling individual variation in growth

An alternative to the approach outlined above, and first introduced to fisheries by Sainsbury (1980), is to assume that each animal grows according to its own growth curve. The basic approach involves developing a pdf for the observed growth increment conditional on the premoult size, the time-at-liberty and the parameters assumed to vary among individuals. For example, under the assumption that the ℓ_∞ parameter (the asymptotic size) varies among individuals (and the growth rate parameter κ is the same for all individuals), this pdf is:

$$p\left(\Delta L_i | l_{1,i}\right) = p\left(\ell_\infty(\Delta L_i) | l_{1,i}\right) \frac{\mathrm{d}}{\mathrm{d}\Delta L} \ell_\infty\left(\Delta L_i\right) \tag{12}$$

where the function $\ell_\infty(\Delta L)$ is selected to account that for the fact that $\ell_\infty - l_1$ is positive (Troynikov, 1988) as well as any dependence of the mean and variance of $\ell_\infty - l_1$ on l_i (Bardos, 2005). This function depends on the assumed growth curve (e.g. Gompertz for Troynikov (1988) and Bardos (2005)). The distribution for $\ell_\infty - l_1$ can take one of many distributions on positive real numbers, including the gamma, lognormal and Weibull (e.g. Troynikov, 1988) Account can also be taken of measurement error when defining the likelihood function which is maximized to estimate the parameter values (Bardos, 2005) and also of the impacts of size-selective fishing mortality, if size-selectivity can be quantified independently of the assessment model (Punt et al., 2006).

Wang et al. (1995) provide an alternative approach to that of Troynikov (1988), Bardos (2005), and Punt et al. (2006) which is based on the joint density of the tag and recapture sizes, i.e.:

$$p\left(l_{1,i}, l_{2,i} | \delta\right) = f\left(A_i\right) g\left(\ell_{\infty,i}\right) \left\|D\right\| \tag{13}$$

where A_i is $1 - \exp\left(-\kappa a_i\right)$, κ is a growth rate (assumed the same for all animals), a_i is the (unknown) age of the ith animal at tagging, $\ell_{\infty,i}$ is the asymptotic size for the ith animal, g and f are pdfs selected based on visual examinations of fits (Wang et al., 1995), and $\left\|D\right\|$ is the determinant of the Jacobian of the transformation

from (l_1, l_2) to (A, ℓ_∞). This formulation relies on the assumption that the age-at-tagging is independent of the asymptotic size. Wang et al. (1995) note that data on animals which were tagged but not recaptured can inform the distribution f and that account can also be taken of measurement error. This basic approach has been extended by several authors. For example, Laslett et al. (2002) generalized the original approach of Wang et al. (1995) to allow $l_2 < l_1$ and for growth to be represented by a growth curve in which the growth rate parameter changes with age. Lloyd-Jones et al. (2014b) developed an approach for calculating size-transition matrices that allows for individual variation in growth parameters and estimates the parameters needed to compute the size-transition matrix using moments.

The methods of Wang et al. (1995), Troynikov (1988), Laslett et al. (2002), Bardos (2005), and Lloyd-Jones et al. (2014a) are all based on the assumption that either ℓ_∞ or κ vary among individuals but not both. Results of simulations by Punt et al. (2009) and Eveson et al. (2007) suggest that there is little value in a method which allows both ℓ_∞ or κ to vary among individuals. Hartm and Chute (2009) implement a method for estimating size-transition matrices allowing for individual variation in ℓ_∞ and κ within a linear mixed-effects modelling framework.

3. Integrating estimation of growth into statistical assessment methods

Traditionally, the size-transition matrix is estimated outside of the model (e.g. Bergh and Johnston, 1992; Chen et al., 2005). For example, Chen et al. (2005) note 'Because of the complexity of lobster growth and lack of tagging information that covers the whole range of lobster size, the growth transition matrix was determined outside the model'. However, with the availability of algorithms for fitting complex models, in particular AD Model Builder (Fournier et al., 2012; Bolker et al., 2013), it has become feasible to include a component in the assessment likelihood function to represent the tag-recapture data. For example, Punt et al. (2010) include tag-recapture data for tiger prawns *Penaeus semisulcatus* and *Penaeus esculentus* in the likelihood function following Eq. (9b), while Haist et al. (2009) include tag-recapture data directly in an assessment for rock lobsters *J. edwardsii* off New Zealand.

One reason for estimating the size-transition matrix in the assessment is so that the other data included in the assessment, in particular size-composition data, can inform growth. Specifically, if the range of sizes included in the tagging experiment was narrow, it is possible for the growth curve to predict numbers-by-size for sizes which are not observed in catch and survey samples (see below). For example, if most tags are placed in animals smaller than the minimum legal size, growth to larger sizes may be relatively uninformed and so over- or under-estimated. This can lead to bias in size-based assessments since the relative numbers of animals surviving to larger sizes is a primary source of information about mortality rate, and thus, of absolute abundance. These errors can be reduced within the assessment by the fitting to size-composition data. The number of parameters added (usually two defining the growth curve and one to characterize variation in growth for each sex) is often only a small fraction of the number of estimated parameters in a typical assessment.

The benefits of integrating estimation of the size-transition matrix into the assessment include the ability for all of the data sources to inform growth and so that that the uncertainty associated with the estimation of growth can be reflected in the measures of uncertainty for key model outputs. However, care needs to be taken when integrating estimation of growth into an assessment to ensure that the values for the parameters of the growth curve and hence the entries of the size-transition matrices are biologically realistic. In addition, the estimates of key model outputs may be biased if some of the pre-specified parameters of the assessment (such as natural mortality) are mis-specified. While it is never possible to eliminate model mis-specification, this can often be detected by examining the residuals about the fit to the tagging data (see below) and through retrospective analyses (e.g. Mohn, 1999; Hurtado Ferro et al., in press).

4. Implications of error for stock assessment

Error due to the size-transition matrix can arise for four main reasons: (a) the model underlying the size-transition matrix is incorrect, (b) the estimation method provides biased estimates, (c) the amount of data is low so the estimates of the entries of the size-transition matrix are imprecise, and (d) growth is estimated within the assessment but one or more biological process (e.g. natural mortality) are mis-specified. There are three main ways to evaluate these errors: (a) sensitivity tests, (b) simulation studies, and (c) comparison of model predictions with data.

4.1. Sensitivity tests

Chang et al. (2012) compare the results of applying various methods for estimating growth curves for crustaceans, and provide model-averaged estimates. Punt et al. (2009) compare the estimated expected growth increment, standard deviation of growth increment and the equilibrium population size distribution for three prawn species for three classes of methods for estimating growth curves and size-transition matrices (allowing for individual variation in growth, assuming that deviations about a mean growth curve are due to observation or process error, and fitting the size-transition matrix directly). There were notable differences among the methods in terms of each of these outputs (e.g. Fig. 1, which show results for four of the methods considered).

Figs. 2 and 3 further explore the sensitivity of the results of assessments to how size-transition matrices are dealt with. Fig. 2 shows fits to survey length-frequency data for three prawn species in Australia's northern prawn fishery (*P. semisulcatus* and *P. esculentus*, and *Metapenaeus endeavouri*) based on the size-structured assessment method of Punt et al. (2010) when growth is estimated in the assessment and when the growth parameters are pre-specified based on analysing the tag-recapture data outside of the assessment. The fits to the length-frequency data are notably poorer when growth is pre-specified. A key reason for the discrepancies in Fig. 2 is the narrow range of tagged prawns compared to those observed in survey and fishery catches. The misfits shown in Fig. 2 translate into differences in key model outputs, notably in the time-trajectory of spawning stock size (Fig. 3).

4.2. Simulation evaluation

Using simulation, Szuwalski and Punt (2012) examined the implications of the number of tag-recaptures available to estimate a size-transition matrix for subsequent use in an assessment of snow crab *Chionoecetes opilio* in the eastern Bering Sea. They found that increasing the number of tag-recaptures from 14 (the number available when the study was conducted) to 100 reduced bias and increased precision for estimates of biomass, B_{MSY}, and the overfished level calculated based on a harvest control rule.

It is well-known (e.g. Sainsbury, 1980; Maller and Deboer, 1988; Wang et al., 1995) that estimates of the parameters of a growth curve are biased if there is individual variation in growth parameters but this ignored. This is illustrated in Fig. 4, which shows distributions for estimates of κ and ℓ_∞ when the true value of κ is 0.2 yr^{-1} and ℓ_∞ is lognormally distributed with mean 100 cm and a pre-specified CV. Each simulated data set was based on 50

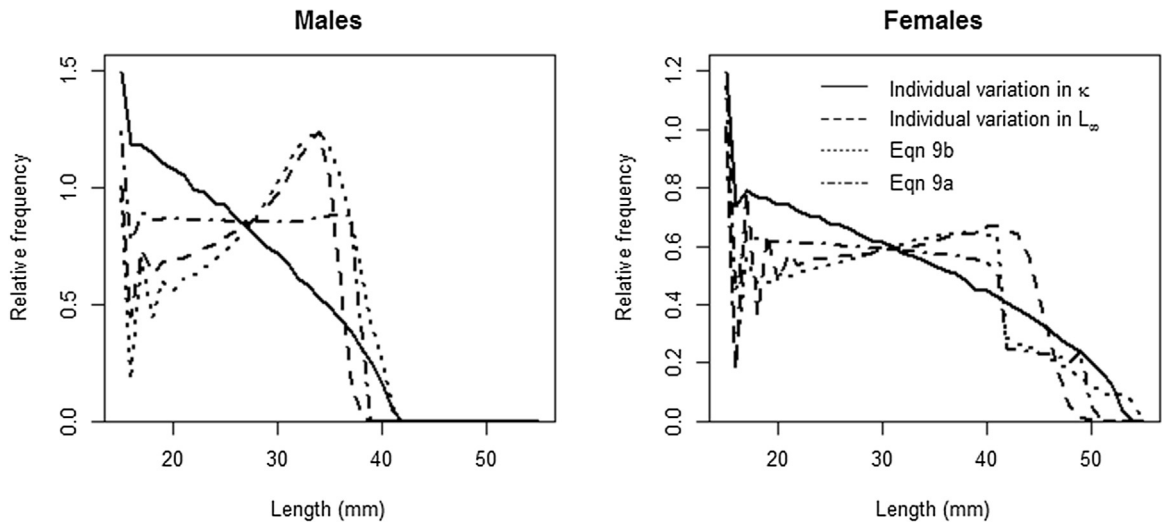


Fig. 1. The equilibrium size composition for *P. semisulcatus* when total mortality is 0.045 week^{-1} for four methods for estimating the size–transition matrix.

individuals, and the time-at-liberty was assumed to be 3 years for all individuals. The extent of bias in the estimates of both κ and ℓ_∞ increases markedly with the CV for ℓ_∞. Fig. 5 shows four example simulated data sets, illustrating the fit to each data set and the bias which arises from individual variation in growth.

Methods which allow for individual variation in parameters tend to lead to unbiased estimates of these parameters (Wang et al., 1995; Punt et al., 2009). However, methods which allow for individual variation in growth parameters are more complex than those which model mean and variance of growth. Consequently, no assessment method exists which integrates the estimation of the size-transition matrix into the assessment when allowance is made for individual variation of growth parameters. Punt et al. (2009) and Lloyd-Jones et al. (2014b) show that size-transition matrices can be constructed based on the assumption that some of the growth curve parameters vary among individuals but the algorithms on which these size-transition matrices are based have also not yet been integrated into assessments.

Punt et al. (2009) conducted the most comprehensive simulation evaluation of the performance of methods for estimating size-transition matrices to date. They compared the performances of eight methods for estimating size-transition matrices, including four methods which are based on the assumption that ℓ_∞, κ, or the age-at-tagging vary among individuals, a method which estimated the size-transition matrix directly, and three methods based on modelling mean growth and variation about that mean. There was no 'best' method. However, Punt et al. (2009) found that methods that allow for individual variation in growth curve parameters performed best, and that methods which assume that ℓ_∞ rather than κ varies among individuals tended to be more robust to violations of model assumptions (see also Eveson et al., 2007). In addition, methods which ignored individual variation in growth predictably

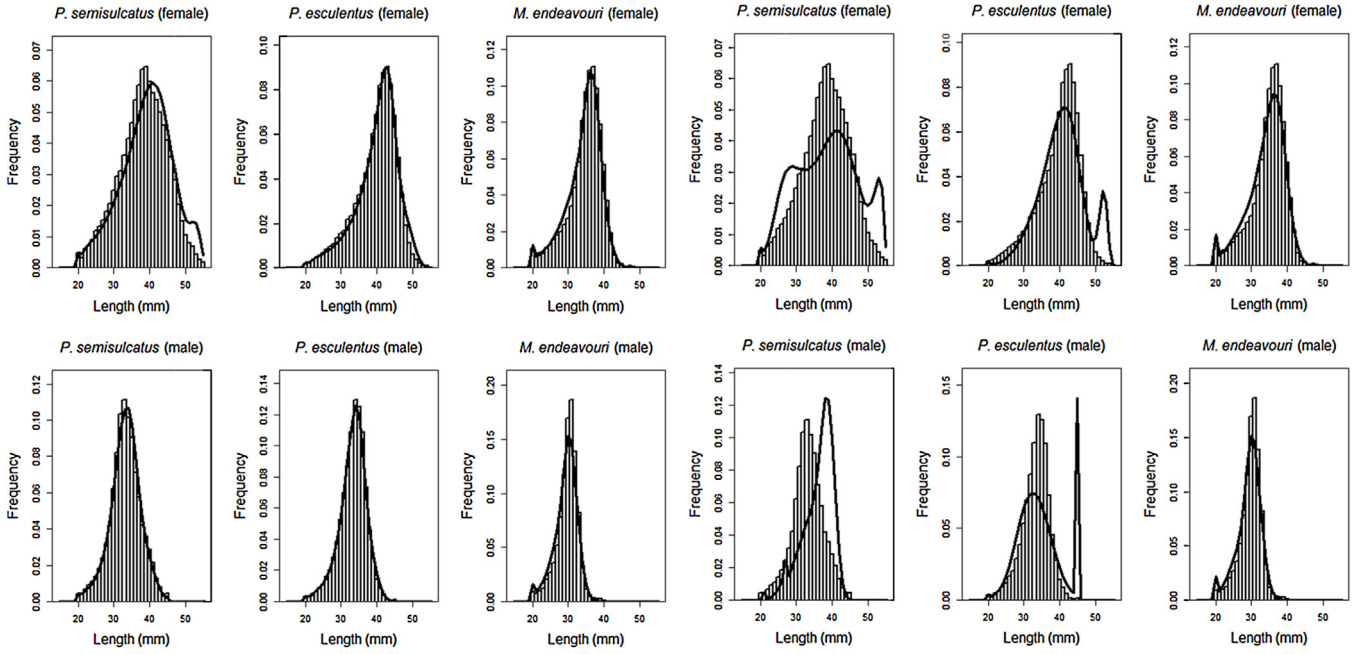


Fig. 2. Observed survey length frequencies (bars) and model predictions from the size-structured population model of Punt et al. (2010) (lines). Values shown are the averages over the years for which data are available (with weights proportional to effective sample sizes). Results are shown when the growth curve is estimated within the model (columns 1–3) and when it is pre-specified based on the published estimates of growth rates (columns 4–6).

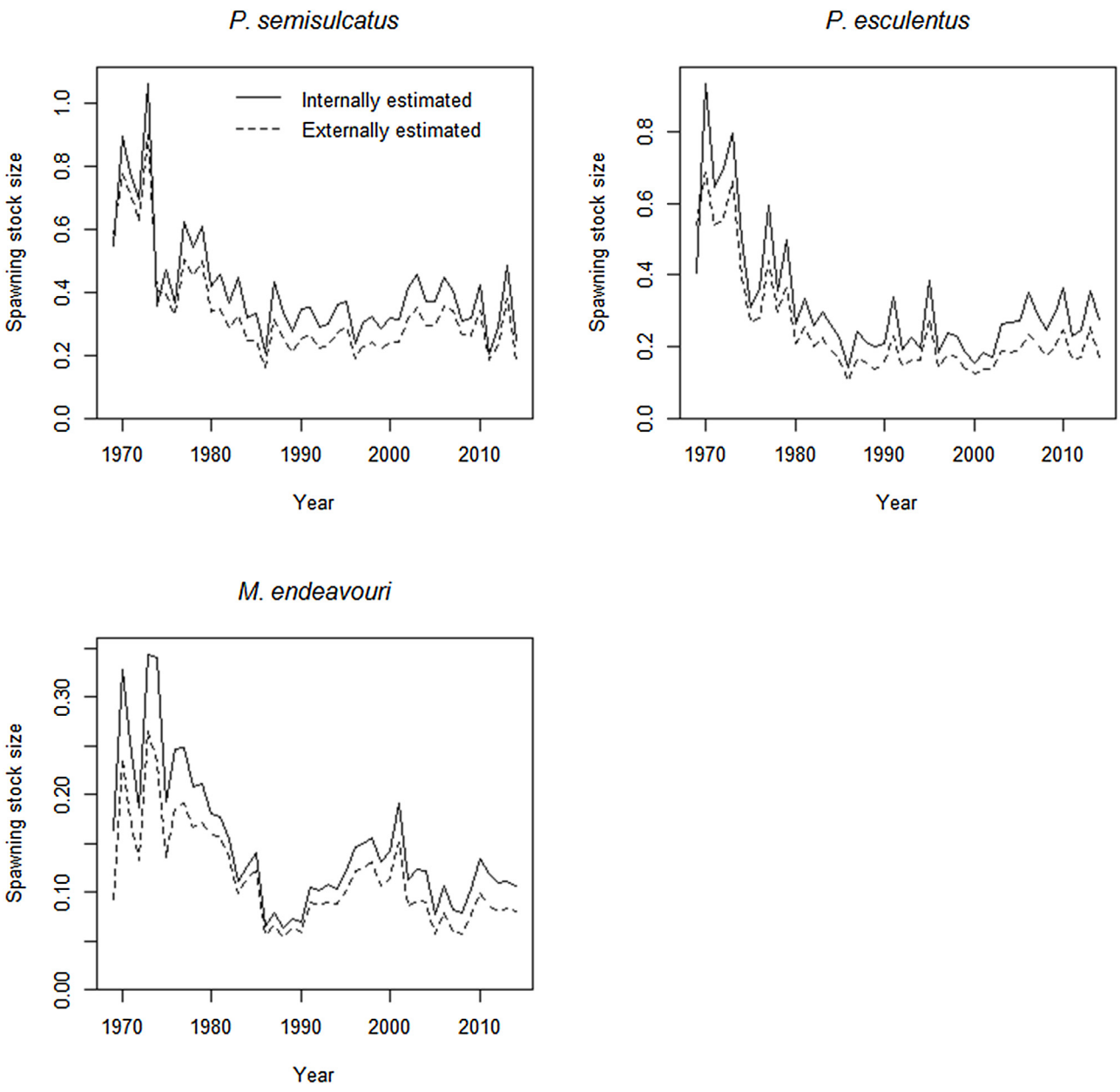


Fig. 3. Time-trajectories of spawning stock size for three species in Australia's prawn fishery. Results are shown when the growth curve is estimated within the model ("Internally estimated") and when it is based on the published estimates of growth rates ("Externally estimated").

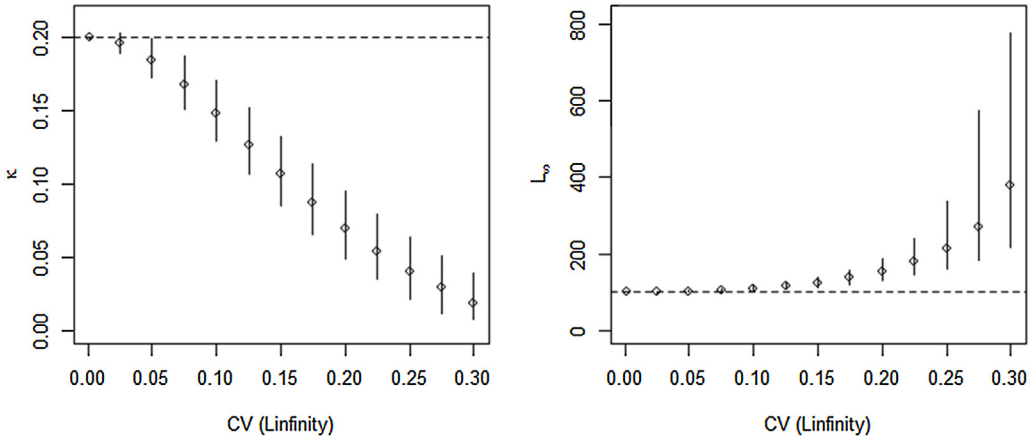


Fig. 4. Distributions of estimates of κ and ℓ_∞ based on simulated tag-recapture data sets in which ℓ_∞ for each individual is sampled from a lognormal distribution. The dotted lines indicate the true value for each parameter (the mean of the true distribution for ℓ_∞).

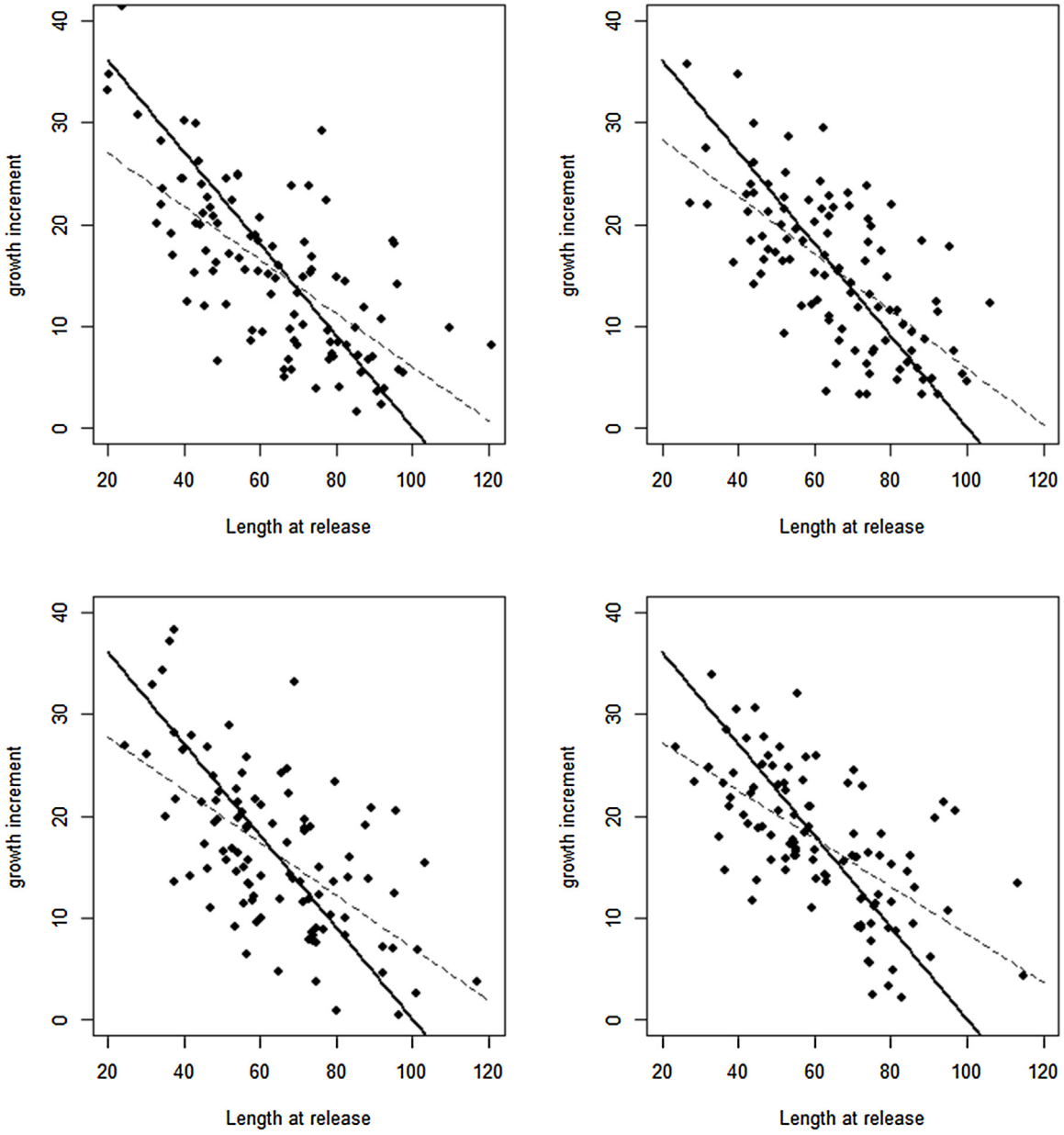


Fig. 5. Four example simulated tag-recapture data sets in which ℓ_∞ for each individual is sampled from a lognormal distribution with a mean of 100 and a CV of 0.15. The solid line is the underlying deterministic relationship and the dashed line is a fit to the generated data.

led to severely biased estimates of population mean values and the size-transition matrix (c.f. Fig. 4).

Punt et al. (2013b) used simulation to evaluate the impact of non-stationarity of various biological and operational parameters (natural mortality, growth, recruitment, catchability) on the performance of the management strategy for rock lobsters *J. edwardsii* off Victoria, Australia when the assessment assumed that these processes were stationary. As expected, violation of the assumption of stationarity led to poor performance. However, allowing growth to change over time in five-year blocks tended to improve performance, albeit at the cost of lower catches when growth is slower than assumed.

4.3. Other

The calculation of size-transition matrices using Eqs. (7) and (9) assumes a distribution for size within the "source" size-class (often a delta distribution around the size-class mid-point). However, size-selective fishing will lead to the distribution of sizes within a size-class changing over time. Table 1 shows the size-transition matrices at carrying capacity and when fishing mortality is 1 yr^{-1}. As expected, growth rates appear higher (e.g. the probability of not growing is lower) when the population is depleted. The impact of this bias on assessment results depends on when the data used to parameterize the size-transition matrix were collected and the change in fishing intensity over time. However, it does suggest that parameterizing a size-transition matrix using tag-recapture data collected when the population was harvested much less or more intensively than at present should be conducted with care. The effect of this bias is expected to be less with smaller size-classes.

Drouineau et al. (2008) explored the implications of choice of time-step and discretization, how individuals are distributed within size-classes, and the probability distribution for growth variability. They found, as expected, that finer temporal resolution

Table 1
Size-transition matrices for a population in which $\kappa = 0.2\ yr^{-1}$, $M = 0.1\ yr^{-1}$, ℓ_∞ is log-normally distributed with mean 100 cm and CV 0.3, and selectivity is a logistic function of length, with lengths of 50% and 95% selectivity at 50 cm and 70 cm, respectively.

	Size-class									
	≤10 cm	≤20 cm	≤30 cm	≤40 cm	≤50 cm	≤60 cm	≤70 cm	≤80 cm	≤90 cm	>90 cm
(a) No fishing mortality										
1	0.094	0	0	0	0	0	0	0	0	0
2	0.906	0.155	0	0	0	0	0	0	0	0
3	0	0.804	0.268	0	0	0	0	0	0	0
4	0	0.04	0.713	0.391	0	0	0	0	0	0
5	0	0	0.02	0.579	0.476	0	0	0	0	0
6	0	0	0	0.03	0.513	0.502	0	0	0	0
7	0	0	0	0	0.011	0.498	0.524	0	0	0
8	0	0	0	0	0	0	0.476	0.622	0	0
9	0	0	0	0	0	0	0	0.378	0.656	0
10	0	0	0	0	0	0	0	0	0.344	1
(b) Fishing mortality of 1 yr^{-1}										
1	0.094	0	0	0	0	0	0	0	0	0
2	0.906	0.155	0	0	0	0	0	0	0	0
3	0	0.804	0.267	0	0	0	0	0	0	0
4	0	0.04	0.713	0.389	0	0	0	0	0	0
5	0	0	0.02	0.579	0.477	0	0	0	0	0
6	0	0	0	0.032	0.509	0.465	0	0	0	0
7	0	0	0	0	0.014	0.535	0.35	0	0	0
8	0	0	0	0	0	0	0.65	0.413	0	0
9	0	0	0	0	0	0	0	0.587	0.389	0
10	0	0	0	0	0	0	0	0	0.611	1

and more size-classes improved accuracy, albeit at greater computational cost and a requirement for more data to parameterize the size-transition matrix. In addition, assuming that the growth increments are gamma-distributed and animals are uniformly distributed across a size-class was found to lead to better estimation performance. It is, however, unclear how robust this conclusion is to the species life history.

Using tagging data to estimate growth will lead to biased results if the tagging process leads to a retardation of growth or sampled sites are not representative of the whole population because of biological or spatial heterogeneity, as tends to occur with abalone (Haddon and Helidoniotis, 2013; Lloyd-Jones et al., 2014a). The bias when growth for blacklip abalone *Haliotis rubra*, from the south-west coast of Tasmania, Australia, is estimated meant that a size-based stock assessment model was incapable of predicting the proportions of larger animals landed by the fishery (Fig. 6). The growth parameters were greatly altered when growth was estimated within the model (Haddon et al., 2013). Such bias can greatly influence model outputs such as the estimated depletion level and the harvest rate (Fig. 6).

The amount of data available to estimate size-transition matrices is generally limited, which makes validation of the estimates of the size-transition matrix using independent data difficult. However, Moser et al. (2002) validated their estimated size-transition matrix by predicting the size distribution of recaptured crabs based on a tagging experiment, the results of which were not used for parameter estimation and using data from captive study. Punt et al. (2010) evaluated the fit of their model to the tag-recapture data, and the extent to which the tag-recapture and length-frequency data are in conflict by plotting fits to length-frequency data and by comparing the observed size distribution at recapture with the expected size distribution based on the model. McGarvey and Feenstra (2001) validated their model of mean growth by comparing mean predicted growth increments from an estimated size-transition matrix over various model time steps, averaging each growth transition probability over all possible growth-transition pathways. Comparisons were made with a fitted continuous mean growth curve (Francis, 1988) and to raw observed mean growth increments, from the tag-captures. Together with likelihood ratio test results, these comparisons permitted selection of the best size-transition polynomial submodel for male and female lobsters.

For analyses which are based on fitting to mean length at recapture, common diagnostic statistics are the residuals plotted against length-at-release, time-at-liberty and expected length-at-recapture (e.g. Montgomery et al., 2009). Each of these plots should exhibit no evidence for patterns of residuals and heterogeneity.

5. Summary of the state of the art

Unfortunately, many stock assessments which use size-transition matrices fail to document how such matrices are estimated. Consequently, this review (Table 2) focused on stocks of crustaceans and molluscs which are assessed in Australia, New Zealand, South Africa and Alaska, the results of which are used for management purposes, either directly to determine stock status or as the basis for management strategy evaluation.

Growth is estimated within the assessment for the bulk of the assessments of the abalone and rock lobster stocks in New Zealand, and this is also the case for some of the crab assessments in Alaska, as well as for the tiger and endeavour prawns in Australia's northern prawn fishery. In contrast, for rock lobster in Australia and South Africa, the size-transition matrices are estimated before the formal assessment is conducted. The number of assessments for which growth is estimated internally to the assessment has increased substantially in recent years; the number of "Externals" in Table 2 would have been much greater had this review been conducted ten years ago. All but one of the assessments which estimated growth within the model included tag-recapture data in their likelihood functions. The remaining stock, Eastern Bering Sea Tanner crab, *Chionoecetes bairdi*, relied on the size-composition data to estimate growth.

Most of the assessments are based on the assumption that growth follows a linear relationship between pre-moult size and growth increment (i.e. the von Bertalanffy growth curve if growth was continuous), although some of the assessments used the more general Schnute form. One assessment, that for rock lobsters *J. edwardsii* off South Australia, was based on the polynomial formulation of McGarvey and Feenstra (2001). The mean and variance of the predicted growth increment distribution, normal or gamma,

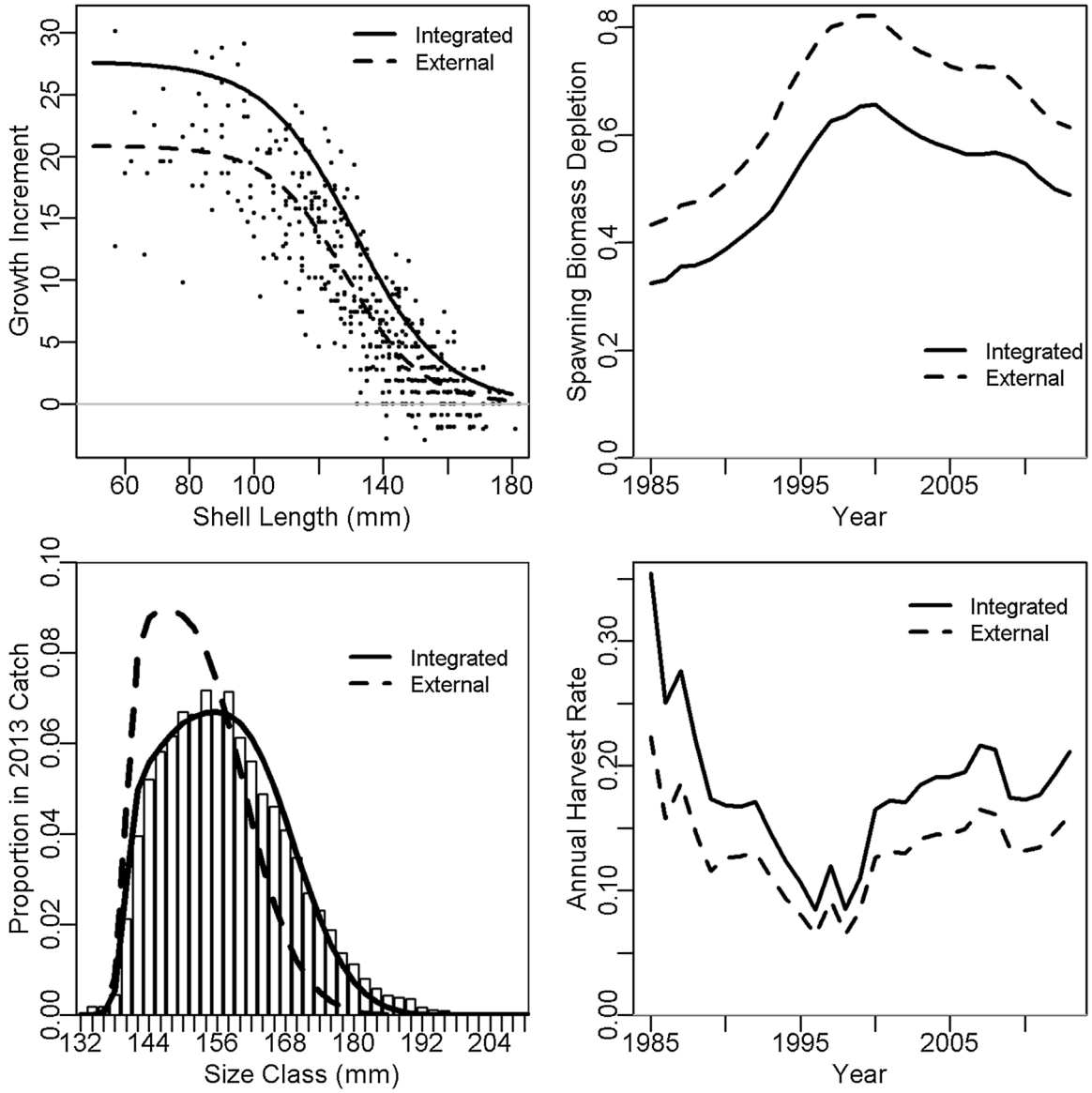


Fig. 6. Outcomes from a size structured stock assessment model applied to blacklip abalone (*Haliotis rubra*) off southwest Tasmania, comparing when the growth transition matrix is estimated externally to the assessment and when it is integrated into the assessment (Haddon, 2014, unpublished) The growth curve used was the inverse logistic (Haddon et al., 2008).

was estimated as polynomial functions of premoult length. This approach permitted a flexible range of growth curves to be represented, including the transition to slower growth typical of female crustacea following maturity. A polynomial of order 2 was sufficient to characterize male lobster growth.

The assessment of *P. camtschaticus* in Norton Sound calculated the mean and variance of the growth increment directly from the tag-recapture data (Zheng et al., 1998). The assessment for Tasmanian giant crab (*Pseudocarcinus gigas*) described growth using tagging data to estimate both the intermoult period and growth increment at size. The size-transition matrix was then estimated using the growth increment estimates, and their variation, while the application of the growth matrix was determined by the intermoult period (McGarvey et al., 2002; Gardner et al., 2007). Most of the assessments either used Eq. (9a) ("Mean" in Table 2) or Eq. (9b) ("Transition" in Table 2) in the estimation of the size-transition matrix. No assessment calculated the size-transition matrix using models which allow for individual variation in growth parameters.

Some of the assessments (those for rock lobsters in Australia) allowed growth to occur multiple times during the year, while only two assessments, those for Bristol Bay and Norton Sound red king crab allowed for time-varying growth.

The assessments in New Zealand impose a minimum amount of variation in growth increment to prevent there being almost no variation in growth for animals close to their estimated asymptotic size.

6. Discussion and next steps

Size-based stock assessments are sensitive to growth estimates, and size-transition matrices typically have many entries. Therefore, the approaches outlined above require substantial amounts of data. Moreover, many abalone and crustacean stocks exhibit spatial structure, in particular, fairly fine scale growth patterns. Consequently, there is often a need to conduct multiple assessments for a single species in a given jurisdiction. For example, the assessment model for rock lobsters off Tasmania accounts for 11

Table 2
Summary of the model structure assumptions on which recent size-structured stock assessments for crustaceans and mollusks off New Zealand, Australia, South Africa and Alaska are based.

Stock	Internal/external	Type of estimation	Functional forms	Additional Notes	Key references
Rock lobster					
Australia					
Tasmania	External	Transition	Schnute; variation in growth normal	Sex-specific; seasonal	Hartmann et al. (2013), Punt et al. (1997)
Victoria	External	Individual	von Bertalanffy; individual variation in κ	Sex-specific	Hobday and Punt (2009)
South Australia	External	Transition	Polynomial mean and variance	Sex-specific	Linnane et al. (2013), McGarvey and Feenstra (2001)
Western Australia	External		Logistic function of size; variation in growth normal with cv 0.05	Sex-specific; seasonal	De Lestang et al. (2011)
New Zealand					
Northland (CRA 1)	Internal	Mean	Von Bertalanffy; variation in growth normal with sd proportional to growth increment; minimum error	Sex-specific	Starr et al. (2003)
Bay of Plenty (CRA 2)	Internal	Mean	Von Bertalanffy; variation in growth normal with sd proportional to growth increment; minimum error	Sex-specific	Starr et al. (2003)
Gisborne (CRA 3)	Internal	Mean	Schnute; variation in growth normal with sd proportional to growth increment; minimum error	Sex-specific	Breen et al. (2009); Starr et al. (2009)
Wairarapa–Hawke Bay (CRA 4)	Internal	Mean	Schnute; variation in growth normal with sd proportional to growth increment; minimum error	Sex-specific	Haist et al. (2009); Breen et al. (2012)
Marlborough Sounds to Cape Jackson (CRA 5)	Internal	Mean	Schnute; variation in growth normal with sd proportional to growth increment; minimum error	Sex-specific; density-dependent	Haist et al. (2009, 2011)
South Africa					
West coast	External	Mean	Adults: linear growth increment with time-varying intercept; juvenile growth (equal for both sexes): quadratic function of length with time blocks. variation in growth normally distributed truncated at +-2sd	Sex-specific; some lobster can shrink; Adults once a year; juveniles multiple times a year, inter-molt period is linear function of size	Brandão et al. (2004) S. Johnston (Univ. of Cape Town, pers. commn)
Abalone					
New Zealand					
Chalky Inlet & South Coast (PAU5A)	Internal	Mean	Exponential; variation in growth normal with sd proportional to growth increment; minimum error	Single-sex	Fu and McKenzie (2010a)
Milford Sound, George Sound, Central Coast & Dusky Inlet (PAU 5A)	Internal	Mean	exponential; error a function of time-at-liberty; minimum error	Single-sex	Fu and McKenzie (2010b)
Stewart Island (PAU 5B)	Internal	Mean	von Bertalanffy; error a function of time-at-liberty; minimum error	Single-sex	Breen et al. (2003); Breen and Smith (2008)
Otago & Southland (PAU 5D)	Internal	Mean	von Bertalanffy; error a function of time-at-liberty; minimum error	Single-sex	Fu (2013a)
Marlborough Sounds, Tasman & Golden Bays (PAU 7)	Internal	Mean	von Bertalanffy; error a function of time-at-liberty; minimum error	Single-sex	Fu (2012)
Southwest Tasmania, (Blocks 9–12)	Internal	Mean	Inverse logistic growth increments; error a function of initial size	Single-sex	Haddon et al. (2008, 2013)
Crab					
Alaska					
Red king crab, Bristol Bay	External	N/A	Von Bertalanffy growth increment pre-specified Time-dependent moult probability for males estimated Time-dependent moult probability for females pre-specified Variation in growth gamma	Sex-specific; probability of maturing estimated; no tagging data in assessment	Zheng and Siddeek (2013)

Table 2 (*Continued*)

Stock	Internal/external	Type of estimation	Functional forms	Additional Notes	Key references
Red king crab, Norton Sound	External	Mean	Mean and variance of growth increment by size-class directly from the tagging data; Time-dependent moulting probability	Male-only model	Hamazaki and Zheng (2013), Zheng et al. (1998)
Red king crab, Pribilof Islands	Internal	Mean	Mean and variance of growth increment by size-class	Male-only model; no tagging data in assessment	Szuwalski et al. (2014)
Snow crab, Eastern Bering Sea	Internal	Mean	von Bertalanffy two-line split at 36.1 mm; female intercept equal to that for males; variation in growth gamma; growth variation pre-specified	Sex-specific All immature moult annually, mature don't moult; terminal moult	Turnock and Rugolo (2013)
Tanner crab, Eastern Bering Sea	Internal	N/A	von Bertalanffy	Sex-specific; no tagging data in assessment; probability of maturing estimated	Stockhausen et al. (2013)
Blue king crab, St Matthews Island	Assumed	N/A		Male-only model	Gaeuman (2013)
Tasmanian giant crab, Australia	External	Mean	von Bertalanffy; moult increments modelled as different logistic regressions for each size class. Moulting from larger size classes takes multiple years.	Sex specific; Numbers at size is a matrix of size class by year, with the number of years being the maximum taken for a size class to fully moult.	Gardner et al. (2007)
Scampi					
Bay of Plenty, Wairarapa/Hawke Bay, and Auckland Islands (SCI 1)	Internal	Mean	von Bertalanffy; error a function of time-at-liberty; minimum error	Sex-specific; seasonal	Tuck and Dunn (2012)
Oyster					
Foveaux Strait, New Zealand	External	Mean	von Bertalanffy; error a function of time-at-liberty; minimum error	Single-sex	Fu (2013b)
Prawns					
Australia's northern prawn	Internal	Transition	von Bertalanffy; variation in growth normal	Sex-specific	Punt et al. (2010)

areas (Hartmann et al., 2013). There are insufficient data to estimate size-transition matrices for all of the 11 areas concerned, so the size-transition matrices on which the assessments are based are either estimated for an area or set to the average of the matrices for adjacent areas (Punt and Kennedy, 1997). In principle, the methods of this paper could be generalized so that parameters are related to location parametrically, which would provide a way to estimate size-transition matrices for all areas using all available data. These could additionally account for variation in population density, which is known to influence growth rates of many invertebrates (e.g. McGarvey et al., 1999) and which will influence optimal levels of exploitation.

The methods of this paper are based on the assumption that growth is homogenous across the range of the assessed stock. This is seldom valid (e.g. McGarvey et al., 1999; Punt et al., 2006). The solution to this problem is to conduct spatial assessments as in Tasmania, but this comes at the cost of correspondingly fewer data to characterise growth in each subregion or the need to integrate growth data spatially (e.g. Chen et al., 2003).

There have been calls to apply model averaging to provide more robust inferences regarding growth (e.g. Katsanevakis and Maravelias, 2008). However, model averaging of size-transition matrices has not yet been accomplished. Given that the entries of a size-transition matrix are comparable across all growth models, model-averaging may be more feasible if the size-transition matrices rather than the growth curves themselves are averaged.

Most growth studies are based on the assumption that animals do not shrink after moulting (Laslett et al. (2002) being a noteworthy exception). Several, but not all studies also ignore recaptured animals which have not been at liberty sufficiently long to become fully mixed into the population. For example, Punt et al. (2010) ignored animals which were at liberty for fewer than two weeks and which were infected (at release or recapture) by the isopod bopyrid parasite *Epipenaeon ingens*, with affects growth. In addition, data for animals which have yet to moult should be ignored if tagging data are to be used to estimate the entries of the matrix which determines the probability of growth to each size-class given that it has moulted.

Finally, although estimation of the standard errors for the parameters of the growth model and hence the size-transition matrix are seldom estimated because the size-transition matrix is assumed to be known exactly when it is computed independently of the assessment, it is possible to estimate these standard errors using asymptotic (e.g. Francis, 1988), Bayesian or bootstrapping methods. In principle, the samples from the latter two approaches could be used as the basis for sensitivity tests. It is also advisable to compare the mean growth increments predicted by size-transition matrices with more direct measures of growth, for different pre-growth body sizes.

Based on this review, we draw the following conclusions and recommendations regarding the future of estimating size-transition matrices.

- Invertebrate growth is known to be highly flexible, responding to variations in population density, maturity status, temperature, habitat, depth, and other environmental and biological influences. Accounting for these, and other sources of variation in growth rates over time and space, may improve

size-based assessments which are sensitive to growth assumptions. However, the ability to account for such variation depends on the availability of relevant data, particularly tagging data, at the appropriate scale and quantity.

- More attention needs to be given to using methods which allow for individual variation in growth parameters as these methods have been shown to perform better in simulation analyses. In particular, these methods should be integrated into current assessment methods. Similarly, because simulation testing has suggested only ℓ_∞ or κ should vary individually, future test cases can assume the true (simulated) variation is in κ while the estimation modelled variation is in ℓ_∞, or vice versa.
- Few of the assessments report detailed diagnostics of the fits to the tagging data when these data were integrated into the assessments and currently objective methods do not exist to weight fits of growth parameters to tagging data, unlike the case for composition and index data (Francis, 2011).
- Research needs to be conduct to better understand the benefits of fewer vs more size-classes in size-structured models as well as how to decide the maximum number of size-class over which growth can occur. Some simulation work has been undertaken to examine the first of these issues for age-structured and size-structured models (e.g. Monnahan et al., 2016; Szuwalski, 2016).
- The ability to estimate size-transition matrices depends on the sampling design for the tagging experiment. However, tagging, particularly for rock lobsters, tends to be of undersize animals, which means that the estimates of growth rates for larger animals are extrapolations based on fits to growth rates for smaller animals (e.g. Fig. 2). There is a need to better integrate tagging studies and how their results are used for assessments.
- Guidelines need to be developed to assist in defining 'best practice' regarding how long animals needed to be at liberty before tagging results can be reliably included in assessments.

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